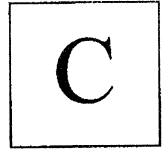


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***B.Tech. Degree V Semester Supplementary Examination
April 2018***

**ME 15-1503 MACHINE TOOLS AND MACHINING SCIENCE
(2015 Scheme)**

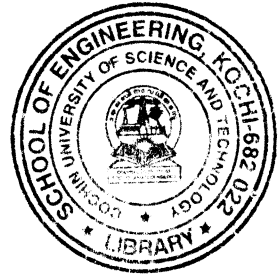
Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) What do you understand by the terms cutting speed and feed in turning operation?
- (b) Differentiate between counter boring and counter sinking.
- (c) How will you specify a grinding wheel? Give an example.
- (d) Differentiate between location and clamping.
- (e) What are the basic mechanisms involved in machining?
- (f) What are the factors favourable for the formation of discontinuous chip during machining?
- (g) Define machinability and machinability index.
- (h) What do you understand by chip reduction coefficient?
- (i) What are the characteristics of ideal cutting tool material?
- (j) Write one method to assess flank wear.



PART B

(4 × 10 = 40)

- II. With a neat sketch explain the following. (10)
 - (i) Steady rest and follower rest
 - (ii) Stepped mandrel
 - (iii) Carriers and catch plates.
- OR**
- III. Explain the crank and slotted link quick return mechanism with a neat sketch. (10)
- IV. (a) Name the abrasives used in grinding wheel. (5)
- (b) Explain the various bonding methods used for making grinding wheel. (5)
- OR**
- V. When will you select V-location? Explain the 3-2-1 principle of location with an example. (10)
- VI. (a) Sketch a cutting tool with the following tool signature 10, 10, 6, 6, 8, 0 (0.8 mm) (5)
- (b) With a neat sketch explain the geometry of a peripheral milling cutter. (5)
- OR**
- VII. Discuss the thermal aspects of machining. (10)

(P.T.O.)

- VIII. (a) Discuss the effect of rake angle in machining. (5)
(b) What are the different causes of tool wear? (5)

OR

- IX. (a) What are the assumptions used for deriving Merchant force relationship? (4)
(b) The following values relate to a cutting test under orthogonal cutting conditions for machining aluminium. Forces as determined by dynamometer FH and FV are (6)

$F_H = 1500 \text{ N}$, $F_V = 1000 \text{ N}$, $\alpha = 10^\circ$, $r = 0.37$

Determine forces along and perpendicular to shear plane and coefficient of friction at tool-chip interface.
